



Stephen Gray transmitted electricity along damp silken thread.



French engineer Henri Pitot devised the Pitot tube to measure the speed of flow beneath the bridges crossing the Seine River in Paris, France.



Catesby's account of flora and fauna

One bird described by Catesby was the Ivory-billed Woodpecker: one of the world's largest woodpeckers and now probably extinct.

These were fused lenses designed to bring different wavelengths of light to focus together.

In the same year, **Joseph**

Foljambe developed a fast, light plow that came to be the standard for the next 180 years. It was called the **Rotherham swing plow** and could be driven by just one man and two horses. The design became so popular it was the first plow ever to be made in a factory.

In North America, British naturalist **Mark Catesby** began to publish the first account of the continent's flora and fauna.

SEVERAL INVENTIONS IN 1731

highlighted the growing scientific interest in measuring the natural world. Italian inventor Nicholas Cirillo created the **first modern seismograph** for measuring the intensity of earthquakes. It consisted of a sensitively balanced pendulum that drew lines on paper as it swung with each tremor, so that the size of the swings recorded their intensity.

Englishman **John Hadley** and American **Thomas Godfrey** independently invented the **octant** to measure the angle of the stars and Sun at sea by lining up their image in a mirror with the horizon. The addition of a telescope in 1759 was important and the octant was widely used for navigation.

In that same year, agricultural changes gained impetus with Jethro Tull's book on horse-hoeing husbandry, which showed

that crops could be sown every year without a fallow period.

Dutch scientist and physician **Herman Boerhaave** put 18th-century chemistry on a firm footing in his book *Elementa Chemiae* (*Elements of Chemistry*), published in 1732. He emphasized meticulous measurement and helped turn chemistry into a science based on principles. Boerhaave also **founded the science of biochemistry** with his brilliant demonstrations on the chemistry of natural substances such as urine and milk.

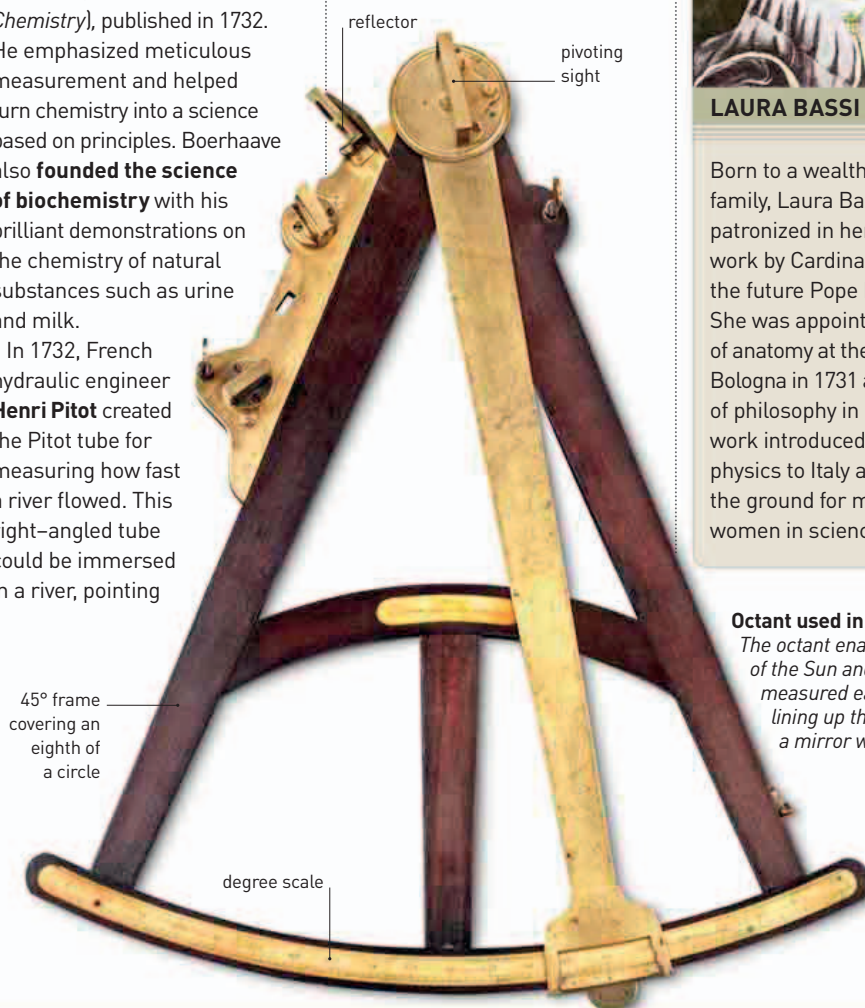
In 1732, French hydraulic engineer **Henri Pitot** created the Pitot tube for measuring how fast a river flowed. This right-angled tube could be immersed in a river, pointing

into the flow so that the height of the water in the upright of the tube indicated the speed of flow. Pitot tubes are now widely used to **measure airspeed on aircraft**.



LAURA BASSI (1711-78)

Born to a wealthy Bolognese family, Laura Bassi was patronized in her scientific work by Cardinal Lambertini, the future Pope Benedict XIV. She was appointed professor of anatomy at the University of Bologna in 1731 and professor of philosophy in 1732. Her work introduced Newtonian physics to Italy and broke the ground for many women in science.



Octant used in navigation

The octant enabled the angle of the Sun and stars to be measured easily at sea by lining up their reflection in a mirror with the horizon.

5 FT/S

THE FLOW VELOCITY OF THE SEINE THROUGH PARIS

